KOMALPREET GREWAL
Sumerpur, Rajasthan 306902
+918209177537 grewalkomalpreet@gmail.com linkedin.com/in/komalpreetgrewal https://github.com/Komalpr3t

Projects

Sherlock Automator – AI Driven Self-Healing Kubernetes Agent | *GitHub* **Nov 2025 - Dec 2025**

- Developed an AI-powered Kubernetes automation agent that detects pod failures through Prometheus monitoring alerts
- Performed automated Root Cause Analysis using local LLMs (Llama 3.2 / TinyLlama) to diagnose infrastructure issues
- Generated automated infrastructure fixes and GitHub pull requests to remediate deployment failures
- Tech used: Kubernetes, Python, Prometheus, Local LLMs (Llama 3.2, TinyLlama), GitHub API

Self-Hosted Generative AI Platform on Kubernetes | *GitHub* **Nov 2025 - Dec 2025**

- Built a local generative AI platform deploying TinyLlama using Ollama on a Kubernetes cluster running on Minikube
- Implemented Open WebUI frontend for interacting with the LLM and backend services exposed through Kubernetes Services
- Configured persistent storage using Kubernetes Persistent Volume Claims to retain model data across pod restarts
- Tech used: Kubernetes, Minikube, Ollama, TinyLlama, Open WebUI, Docker, YAML

End-to-End DevOps E-Commerce Deployment using GitOps | *GitHub* **Oct 2025 - Nov 2025**

- Implemented a complete DevOps lifecycle for a Node.js e-commerce application using containerization and Kubernetes orchestration.
- Provisioned AWS infrastructure including VPC, networking, and EKS cluster using Terraform Infrastructure as Code.
- Automated CI/CD pipeline using GitHub Actions for image builds and ArgoCD GitOps for continuous deployment to Kubernetes.
- Tech used: AWS (EKS), Terraform, Docker, Kubernetes, GitHub Actions, ArgoCD, Node.js

Training

Cipher Schools: Data Structures and Algorithms using Java **June 2025 – July 2025**

- Completed structured DSA training in Java, covering arrays, linked lists, stacks, queues, trees and graphs
- Practiced algorithmic problem-solving using recursion, sorting, searching and dynamic programming techniques
- Gained hands-on experience implementing DSA solutions in Java, improving coding efficiency, time-space complexity analysis and technical interview preparation

Certificates

- Master Generative AI | Infosys Springboard | Link **Aug 2025**
- Build Generative AI Apps | Infosys Springboard | Link **Aug 2025**
- Microsoft Azure Fundamentals | Microsoft | Link **Jan 2025**

Skills

Languages: Java, Python, C, C++, Bash
Technologies: Git, Docker, Kubernetes, AWS (EKS, EC2, VPC), Terraform, ArgoCD, GitHub Actions, Prometheus, Ollama, Open WebUI, Linux
Domain Skills: DevOps, Cloud Computing, Kubernetes Infrastructure, AI Infrastructure Deployment, Data Structures and Algorithms, Problem Solving

Education

Lovely Professional University **Phagwara, Punjab**
Bachelor of Technology - Computer Science and Engineering — CGPA: 8.18 *Aug 2023 – present*

HC Bafna Sr. Sec. School **Sumerpur, Rajasthan**
12th — Percentage: 86.50% *April 2022 – March 2023*

HC Bafna Sr. Sec. School **Sumerpur, Rajasthan**
10th — Percentage: 98.2% *April 2020 – March 2021*